Please come back at 9 AM

[with lots of questions]

$a_1, a_2, \ldots, a_{100}$

Fourier Series Representation of Continuous-time Periodic Signals

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aperiodic ?

F. Transform

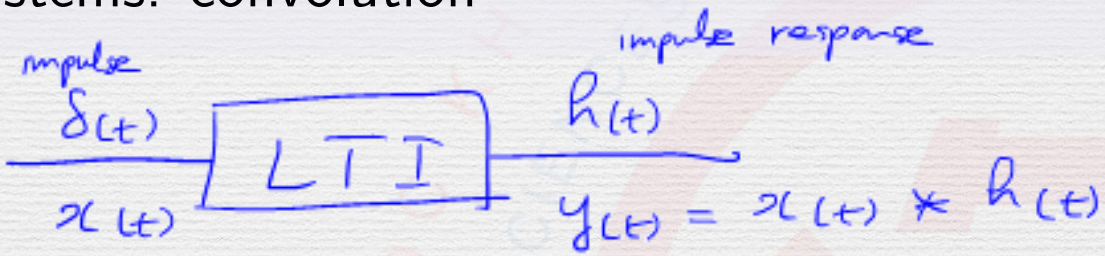
F. Series

↳ Friday

The Course So Far

Until Today

- Signals and Systems
- LTI systems: convolution



The Course So Far

Until Today

- Signals and Systems
- LTI systems: convolution

Today

Fourier Series in Continuous-time domain

Objectives

Background

Fourier Representation

Properties

Homework

1 Lesson Objectives

2 Review Complex Numbers

3 Fourier Representation

4 Properties

5 Homework



1 Lesson Objectives

2 Review Complex Numbers

3 Fourier Representation

4 Properties

5 Homework

Lesson Objectives

At the end of this lesson, you should be able to

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At the end of this lesson, you should be able to

1. be familiar with complex numbers

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At the end of this lesson, you should be able to

1. be familiar with complex numbers
2. find Fourier coefficients of any continuous periodic signals

Lesson Objectives

At the end of this lesson, you should be able to

1. be familiar with complex numbers
2. find Fourier coefficients of any continuous periodic signals
3. understand various properties of Fourier Series

1 Lesson Objectives

2 Review Complex Numbers

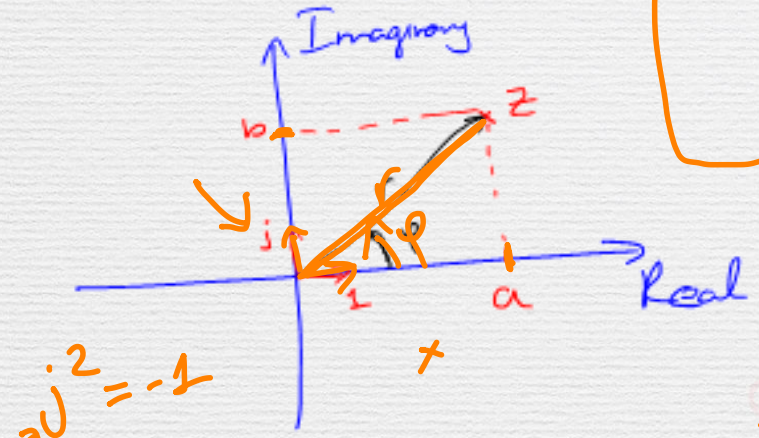
3 Fourier Representation

4 Properties

5 Homework

Review: Complex Number, Conjugation (2,3)

Complex plane with polar and Cartesian representation



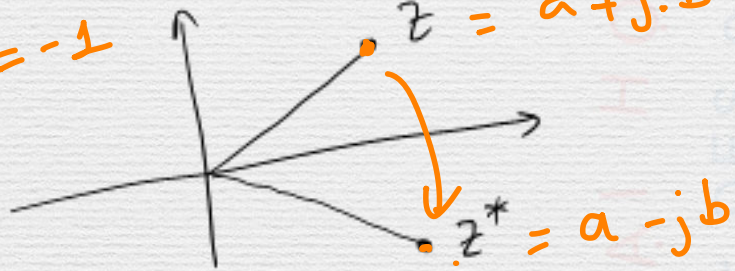
$z = a + j.b \sim (a, b)$

$z = (r, \varphi) \quad (3, \frac{\pi}{4})$

$= r.e^{j\varphi}$

$j^2 = -1$

$i^2 = -1$

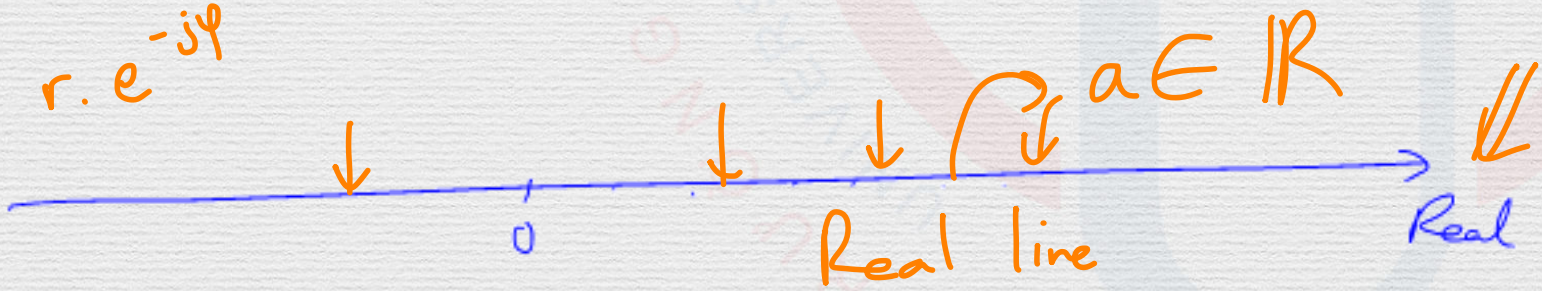


$r.e^{j\varphi}$ $r.7^{j\varphi}$ ...

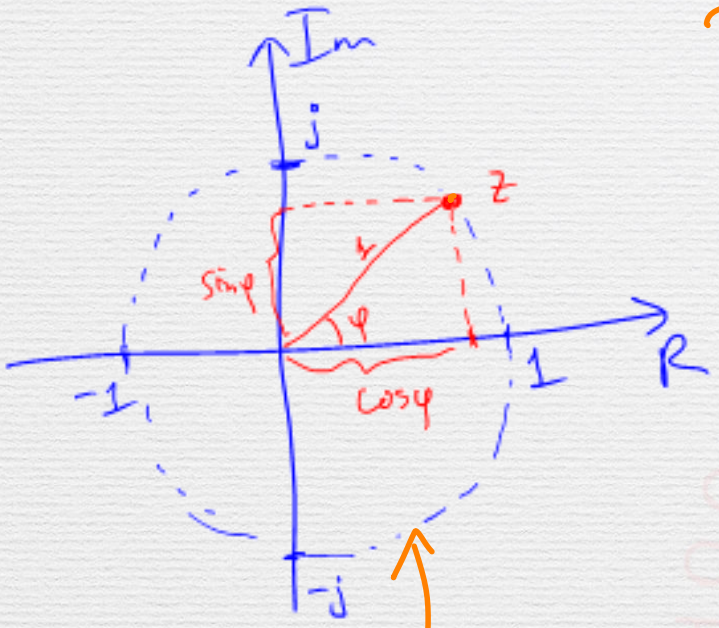
$\frac{r.e^{j\varphi}}{?}$

$r.e^{j\varphi}$

$r.e^{-j\varphi}$



Review: Euler Formula $e^{j\psi} = cos(\psi) + jsin(\psi)$



unit circle

root

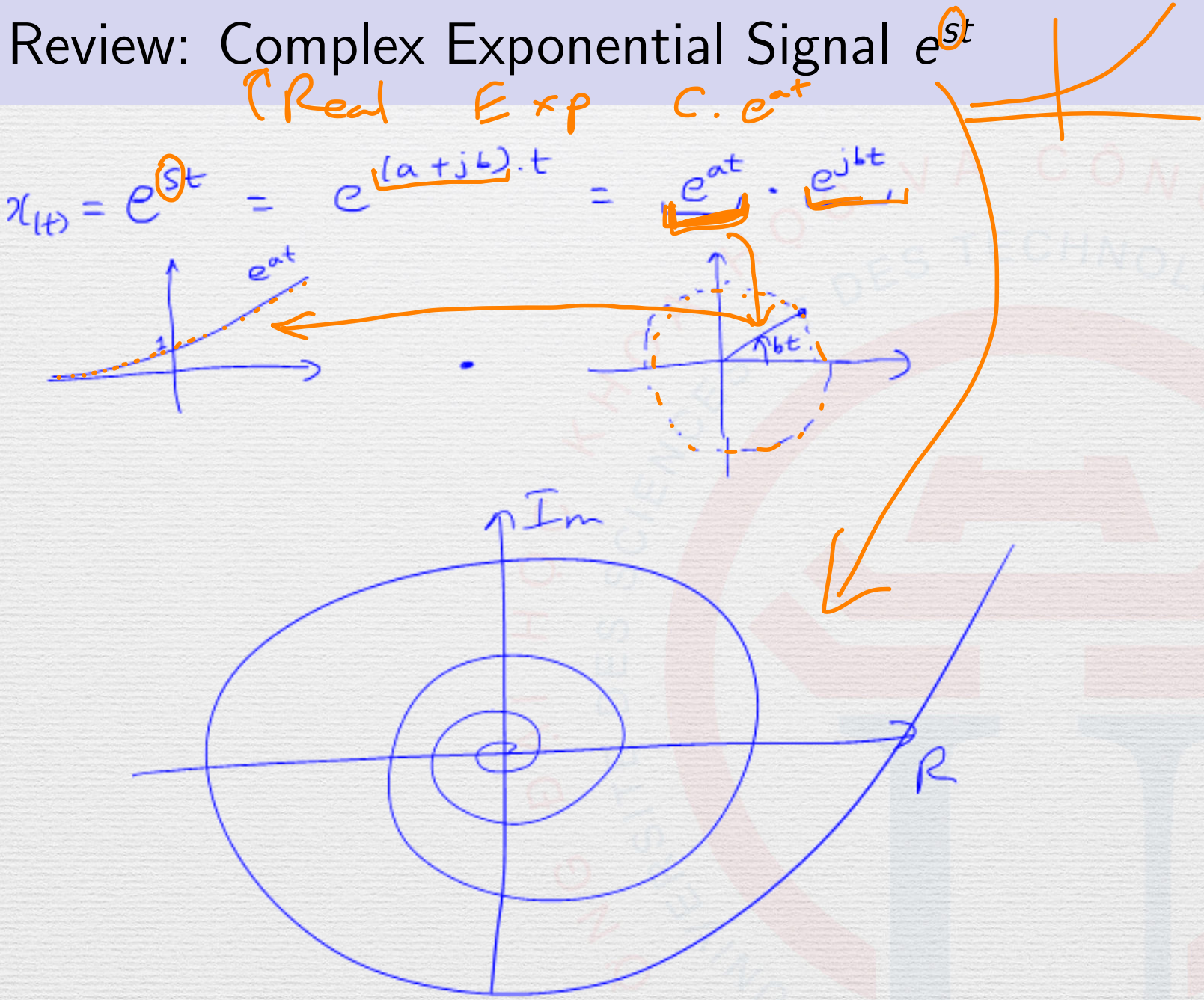
$z = e^{j\varphi} = \cos\varphi + j \cdot \sin\varphi$

$e^{-j\varphi} = \cos\varphi - j \cdot \sin\varphi$

$$\cos\varphi = \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

$$\sin\varphi = \frac{e^{j\varphi} - e^{-j\varphi}}{2j}$$

Review: Complex Exponential Signal e^{st}



LTI Response to Complex Exponential

Convolution Integral

$$y(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

$x(t) = e^{st}$ → $h(t)$ → $y(t)$

$$y(t) = \int_{-\infty}^{+\infty} h(\tau) \cdot e^{st} \, e^{-s\tau} \, d\tau = e^{st} \left[\int_{-\infty}^{+\infty} h(\tau) \cdot e^{-s\tau} \, d\tau \right]$$

$s = \sigma + j\omega$

$$= x(t) \cdot |H(s)|$$

ALL LTI Systems

LTI Response to Complex Exponential

Convolution Integral

$$y(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau$$

The response of an LTI system to a complex exponential input is the same complex exponential with only a change in amplitude.

Objectives

Background

Fourier Representation

Periodic Signals

Definition

Properties

Homework

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3 Fourier Representation
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4 Properties

5 Homework

Fundamental Period

Signals & Systems

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Objectives

Background

Fourier Representation

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Definition

Properties

Homework

The minimum positive, nonzero value of T so that

$$x(t) = x(t+T)$$

Fundamental frequency $\omega_0 = \frac{2\pi}{T}$

2 Basic Periodic Signals

Signals & Systems

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Objectives

Background

Fourier Representation

Periodic Signals

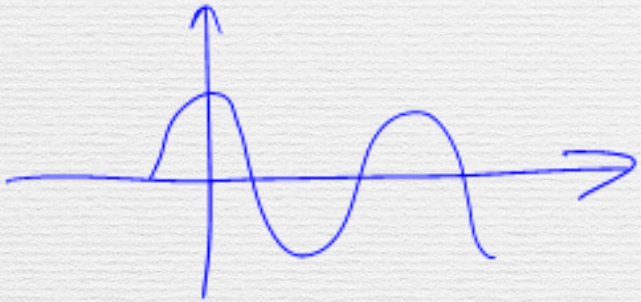
Definition

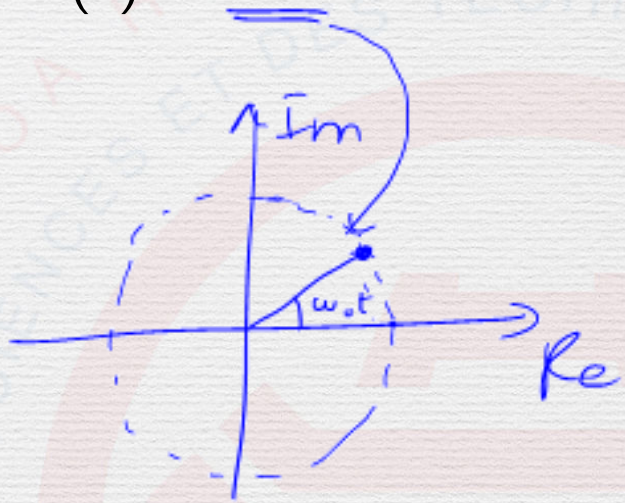
Properties

Homework

the sinusoidal signal $x(t) = cos\omega_0 t$

the periodic complex exponential $x(t) = e^{j\omega_0 t}$





an Example

Consider a periodic signal $x(t)$ with fundamental frequency 2π

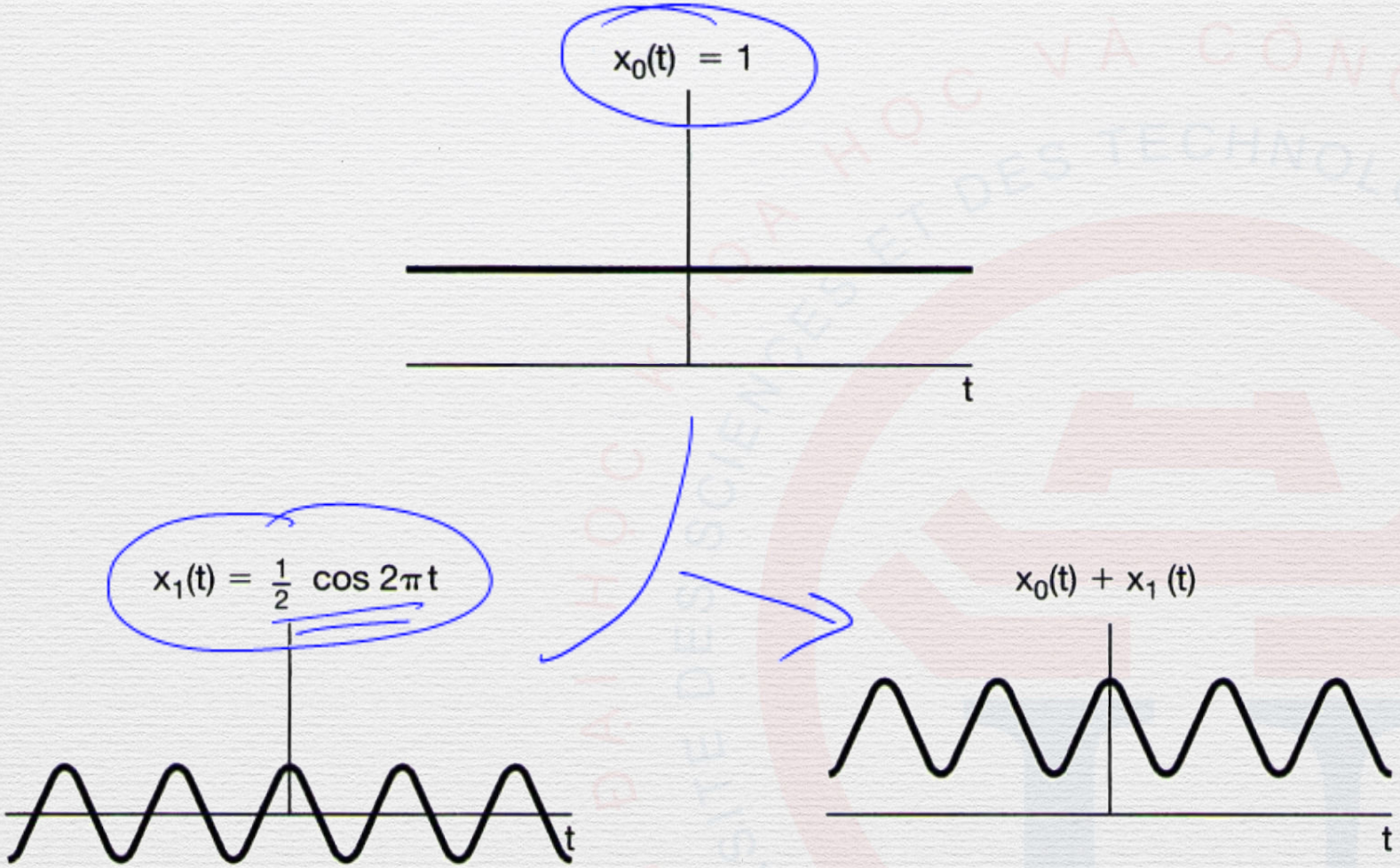
$$x(t) = \sum_{k=-3}^{3} a_k e^{jk2\pi t} \quad = \quad \cos \ldots$$

where $a_0 = 1, a_1 = a_{-1} = \frac{1}{4}, a_2 = a_{-2} = \frac{1}{2}, a_3 = a_{-3} = \frac{1}{3}$

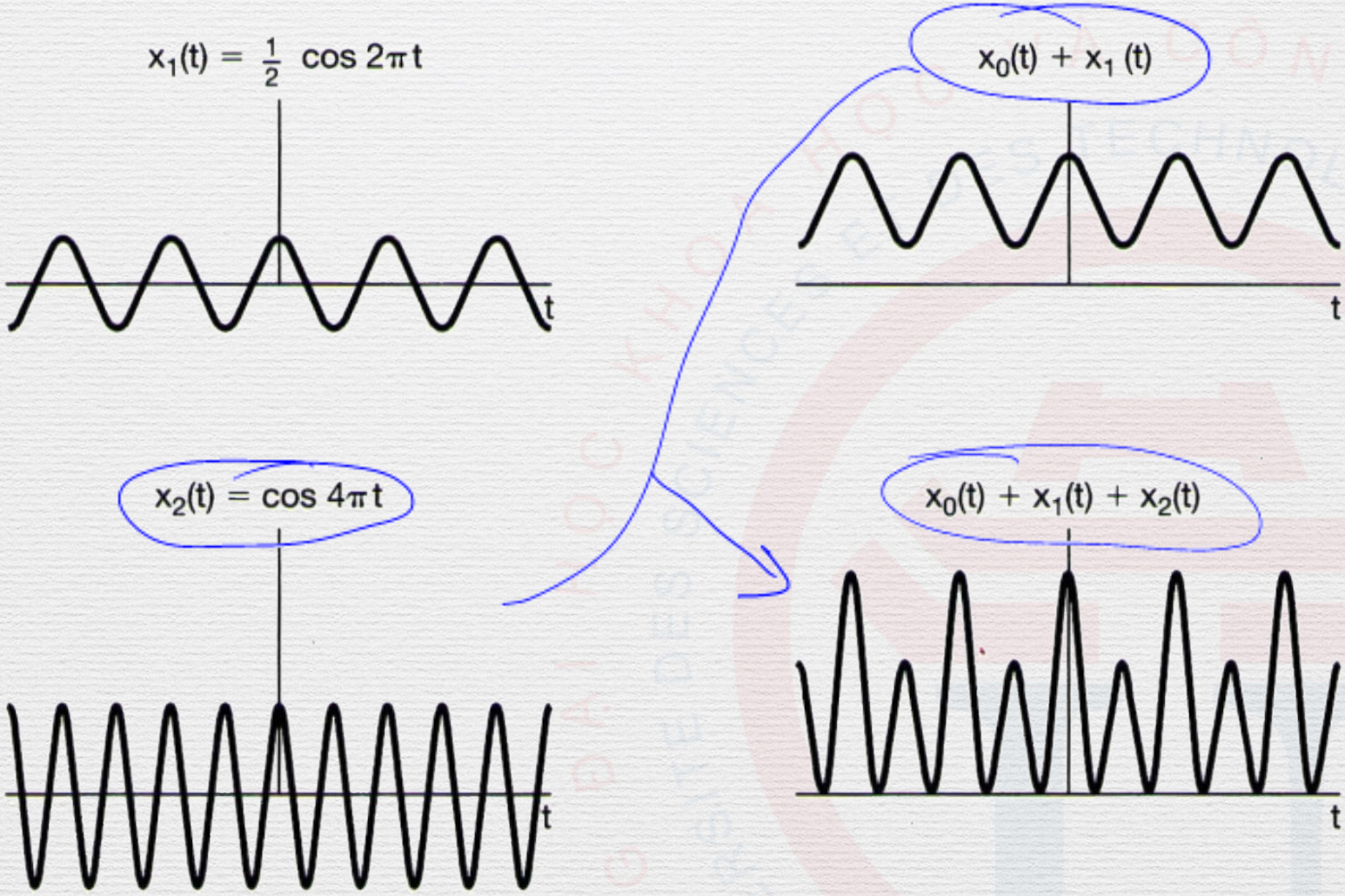
$$\begin{cases} e^{j\varphi} = \cos\varphi + j\sin\varphi \\ e^{-j\varphi} = \cos\varphi - j\sin\varphi \end{cases}$$

$$1 + \frac{1}{2}\cos(2\pi t) + \cos(4\pi t) + \frac{2}{3}\cdot\cos(6\pi t)$$

an Example



an Example



an Example

Signals & Systems

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Objectives

Background

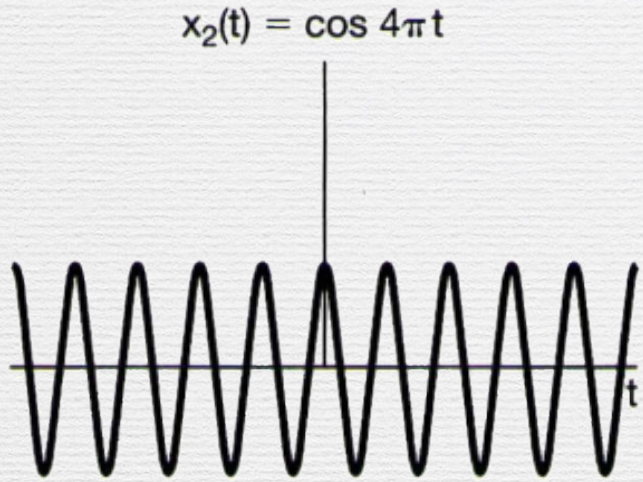
Fourier Representation

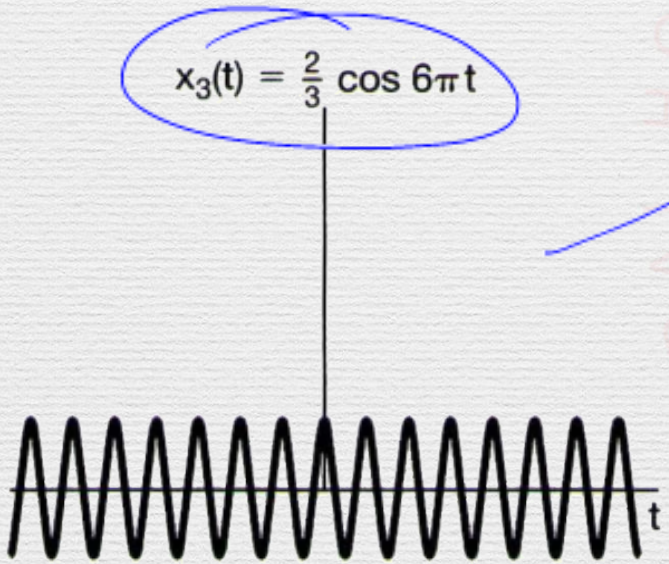
Periodic Signals

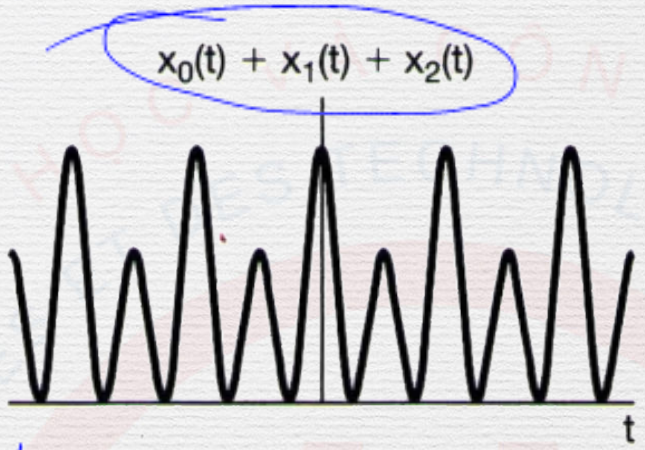
Definition

Properties

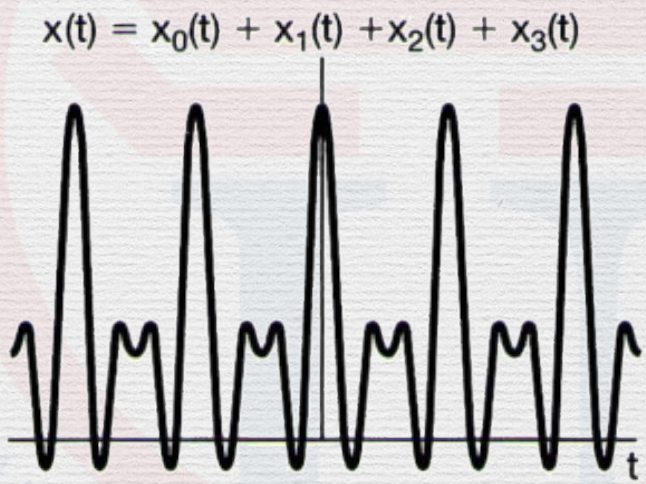
Homework







$$\sum_{k=-3}^{3} a_k \cdot e^{jk\,2\pi t}$$



Review: Integral of $e^{j\omega_0 t}$

Given $\omega_0 = \frac{2\pi}{T}$, calculate

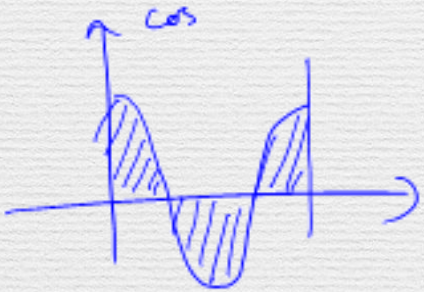
1

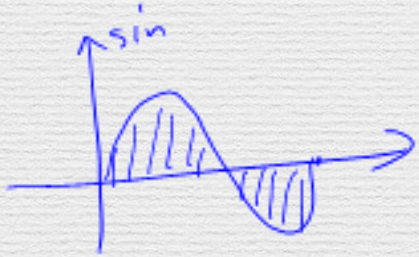
$$\int_0^T e^{j\omega_0 t} dt = 0$$

$$\int_0^T (\cos \omega_0 t + j \sin \omega_0 t)\, dt$$

$\cos \omega_0 t \rightarrow 0$

$\sin \omega_0 t \rightarrow 0$





Review: Integral of $e^{j\omega_0 t}$

Given $\omega_0 = \frac{2\pi}{T}$, calculate

1. $$\int_0^T e^{j\omega_0 t} dt$$

2. $$\int_0^T e^{jk\omega_0 t} dt = \begin{cases} T & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$$

$k = 0 \rightarrow 1 \rightarrow T$

Definition

Assume that

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Then

$$a_k = ??? \quad \frac{1}{T} \int_0^T x(t) \cdot e^{-jk\omega_0 t} dt$$

Hint: calculate $x(t)e^{-jn\omega_0 t}$, and its integral from 0 to T.

$$\int_0^T x(t)\, e^{-jn\omega_0 t} dt = \int_0^T \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \cdot e^{-jn\omega_0 t} dt$$

$$= \sum_{k=-\infty}^{+\infty} a_k \int_0^T e^{j(k-n)\omega_0 t} dt \doteq a_n \cdot T$$

$$\begin{cases} T & \text{if } k = n \\ 0 & \text{else} \end{cases}$$

Fourier Series Equation Pair

periodic signals : T

$\omega_0 = \frac{2\pi}{T}$

Definition

The synthesis equation

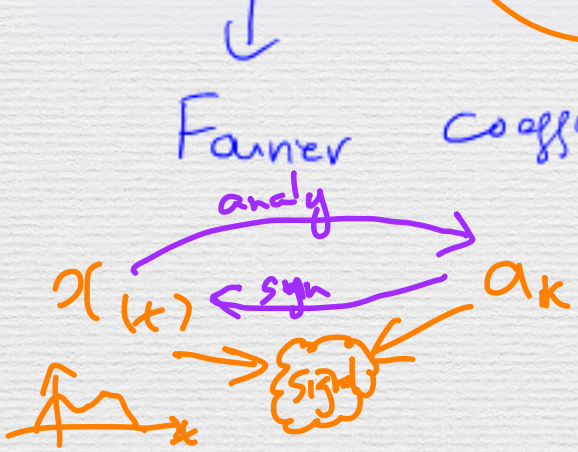
$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk\frac{2\pi}{T}t}$$

The analysis equation

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk\frac{2\pi}{T}t}dt$$

Fourier analy

coefficients



$|[\ldots]| = T$

$\int_0^T$ $\int_{-T/2}^{T/2}$ $\int_{6T}^{7T}$ $\int_{-5T}^{-4T}$

Example 1

Example

Determine Fourier series coefficients of

$$x(t) = sin\omega_0 t$$

$$= \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$a_1 = \frac{1}{2j} \qquad a_0 = a_2 = a_{\pm 2} = \cdots = 0$$

$$a_{-1} = -\frac{1}{2j}$$

Example 2

Signals & Systems

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Objectives

Background

Fourier Representation

Periodic Signals

Definition

Properties

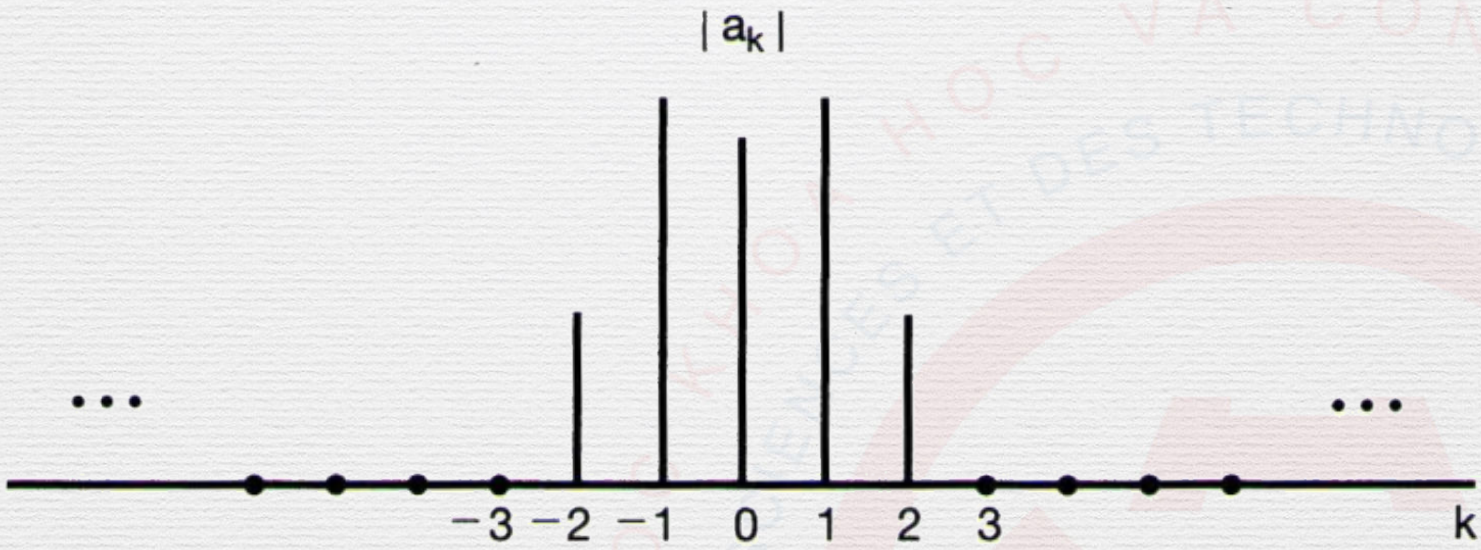
Homework

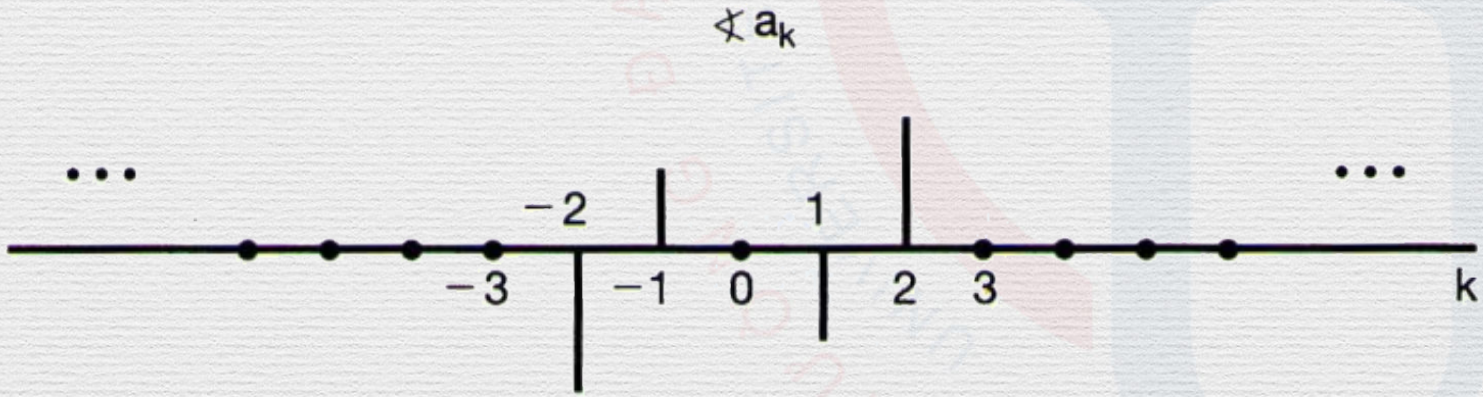
Example

Determine Fourier series coefficients of

$$x(t) = 1 + sin\omega_0 t + 2cos\omega_0 t + cos(2\omega_0 t + \frac{\pi}{4})$$

Example 2



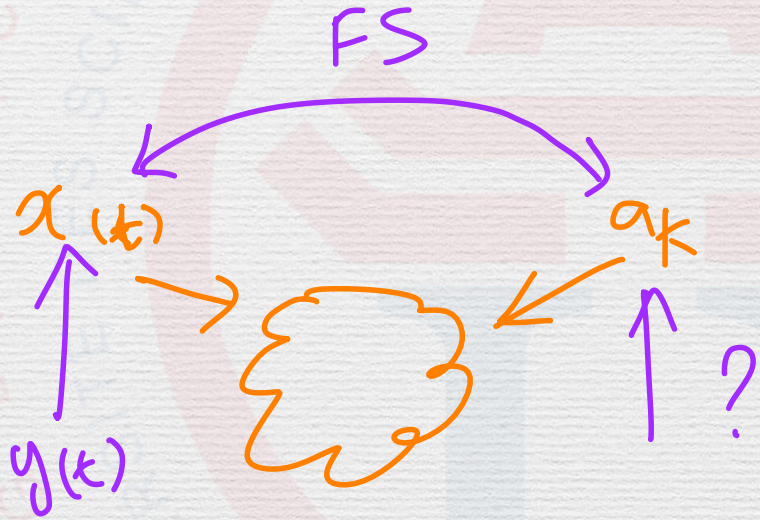


1 Lesson Objectives

2 Review Complex Numbers

3 Fourier Representation

4 Properties
- Linearity
- Time Shifting
- Time Reversal
- Time Scaling
- Multiplication
- Conjugation
- Parseval's Relation



5 Homework

Notations

$x(t)$ is a periodic signal with period T and fundamental frequency $\omega_0 = \frac{2\pi}{T}$. Its Fourier series coefficients are denoted by a_k:

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Fourier Pair

The synthesis equation

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk\frac{2\pi}{T}t}$$

The analysis equation

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk\frac{2\pi}{T}t}dt$$

Linearity

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

$$y(t) \overset{\mathcal{FS}}{\longleftrightarrow} b_k$$

Then

$$z(t) = Ax(t) + By(t) \overset{\mathcal{FS}}{\longleftrightarrow} c_k = \ ?$$

Linearity

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

$$y(t) \overset{\mathcal{FS}}{\longleftrightarrow} b_k$$

Then

$$z(t) = Ax(t) + By(t) \overset{\mathcal{FS}}{\longleftrightarrow} c_k = Aa_k + Bb_k$$

Verify this property.

Time Shifting

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x(t - t_0) \overset{\mathcal{FS}}{\longleftrightarrow} C_k$$

Time Shifting

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x(t - t_0) \overset{\mathcal{FS}}{\longleftrightarrow} e^{-jk\omega_0 t_0} a_k$$

Verify this property.

Time Reversal

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x(-t) \overset{\mathcal{FS}}{\longleftrightarrow}$$

Time Reversal

Signals & Systems

TRAN Hoang Tung

Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x(-t) \overset{\mathcal{FS}}{\longleftrightarrow} a_{-k}$$

!!!

Verify this property.

Time Scaling

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x(\alpha t) \overset{\mathcal{FS}}{\longleftrightarrow} ???$$

Time Scaling

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \xleftrightarrow{\mathcal{FS}} a_k$$

Then

$$x(\alpha t) \xleftrightarrow{\mathcal{FS}} ?$$

Attention:

1. the Fourier coefficients have not changed
2. the Fourier series representation has changed

Multiplication

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

$$y(t) \overset{\mathcal{FS}}{\longleftrightarrow} b_k$$

Then

$$z(t) = x(t)y(t) \overset{\mathcal{FS}}{\longleftrightarrow}$$

Multiplication

Signals & Systems

TRAN Hoang Tung

Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

$$y(t) \overset{\mathcal{FS}}{\longleftrightarrow} b_k$$

Then

$$z(t) = x(t)y(t) \overset{\mathcal{FS}}{\longleftrightarrow} c_k = \sum_{l=-\infty}^{+\infty} a_l b_{k-l}$$

$a_k * b_k$

Verify this!

Conjugation

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x^*(t) \overset{\mathcal{FS}}{\longleftrightarrow}$$

Conjugation

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

Given

$$x(t) \overset{\mathcal{FS}}{\longleftrightarrow} a_k$$

Then

$$x^*(t) \overset{\mathcal{FS}}{\longleftrightarrow} a^*_{-k}$$

Parseval's Relation

Signals & Systems

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Objectives

Background

Fourier Representation

Properties

Linearity

Time Shifting

Time Reversal

Time Scaling

Multiplication

Conjugation

Parseval's Relation

Homework

Definition

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

Power

Objectives

Background

Fourier Representation

Properties

Homework

1 Lesson Objectives

2 Review Complex Numbers

3 Fourier Representation

4 Properties

5 Homework

Homework

Continuous Fourier Series Exercises

3.1, 3.4, 3.5, 3.17, 3.22, 3.46